

Diversity of Munc13 and PSD-95 Clusters in Excitatory Synapses onto Different Inhibitory Neuron Subtypes

Diya Sharma
Department of Physiology
University of Maryland, School of
Medicine
Baltimore, United States of America
DiyaSharma@som.umaryland.edu

Aaron Levy
Department of Physiology
University of Maryland, School of
Medicine
Baltimore, United States of America
Aaron.Levy@som.umaryland.edu

Thomas A. Blanpied
Department of Physiology
University of Maryland, School of
Medicine
Baltimore, United States of America
TBlanpied@som.umaryland.edu

Abstract— Excitatory and inhibitory neurons possess different synaptic structures and molecular characteristics, due to their distinct roles in neural circuitry. Additionally, there exist various subtypes of interneurons with well-defined functional differences. However, it remains unclear whether these functional differences are manifested in the molecular organization of protein clusters at excitatory synapses onto these interneurons. We combine confocal microscopy, quantitative image analysis, and machine learning to investigate synaptic protein cluster organization at different synapse types, focusing on Munc13 and PSD-95 protein clusters at excitatory-to-inhibitory (Ex → Inh) and excitatory-to-excitatory (Ex → Ex) synapses. Furthermore, we employ five neuropeptide markers (PV, SST, CCK, NPY, and VIP) to identify and differentiate subtypes within Ex → Inh synapses, aiming to elucidate finer distinctions in synaptic organization. Our findings reveal larger and denser Munc13 clusters at Ex → Inh synapses compared to Ex → Ex synapses, indicated by a significant increase in cluster area and intensity. Among Ex → Inh synapse subtypes, there are notable differences in the ratios of Munc13 and PSD-95 intensities in certain cases, while in others, these ratios are similar, implying comparable cluster organization among those subtypes. Through dimensionality reduction and clustering techniques, we reveal a continuous spectrum of variation among synapse classes, underscoring the presence of both distinct and shared characteristics across synapse subtypes. These variations play a crucial role in synaptic transmission and correspond to the functional differences among different neuron types. Identifying the correlation between protein organization and synapse subtype establishes a foundation for future experiments aimed at understanding nanocluster organization and trans-synaptic alignment at such synapse types.

Keywords— Munc13, PSD-95, excitatory synapses, inhibitory subtypes, synapse organization, synapse types, fluorescence imaging, unsupervised machine learning

I. INTRODUCTION

Synapses are highly complex structures that involve thousands of different proteins at different amounts and locations. Typically, synapse types are categorized based on the neurotransmitter molecules they release, such as glutamate or GABA. However, within these broad categories, there exist synaptic subtypes that exhibit unique protein profiles and specialized functions within neural circuits [1]. These differences in protein composition and function are crucial because they can influence how information is processed and transmitted in the brain. Additionally, many neurological disorders are associated with mutations in synaptic proteins expressed specifically in certain synapse subtypes [2-4]. Therefore, understanding the diversity of synaptic architecture among various synapse types and being able to classify

synapses on a molecular scale are essential, providing valuable insights into how synaptic diversity contributes to brain function and dysfunction.

The function and efficacy of synapses depend heavily on the precise organization of key cluster-forming proteins. For example, presynaptic vesicle docking proteins, such as Munc13, influence synaptic strength by regulating the number of release sites, release probability, and other features of neurotransmitter vesicle release [5,6]. Postsynaptic scaffolding proteins, such as PSD-95, affect synaptic strength by controlling the number and type of receptors to be activated [7,8]. The organization of these protein clusters not only facilitates neurotransmitter release but also shapes synaptic transmission and plasticity, contributing to distinct characteristics in different cell types and projections. Understanding the molecular changes in the organization of such key proteins is therefore essential. These insights into synaptic protein dynamics not only deepen our understanding of synaptic function but also provide a foundation for addressing broader neurological issues at a macroscopic level.

Advanced techniques like super-resolution microscopy and machine learning have enhanced our understanding of synapse diversity and allowed for the synapse subtype classification. Increasing evidence confirms the existence of excitatory synapse subtypes and highlights differences in protein organization among these subtypes. For example, a combination of multiplexed fluorescence imaging of protein intensities and unsupervised clustering based on synaptic characteristics identified six unique clusters, comprising five excitatory synapse subtypes [9]. Hierarchical clustering of protein intensity profiles and correlation coefficients revealed distinguishing features between excitatory synaptic clusters, including levels of synaptic protein expression, proportions of synapses with undetectable protein levels, and patterns of coordinated protein expression.

In addition to protein profiles, changes in synaptic architecture can be correlated with postsynaptic cell type. In the hippocampus, excitatory synapses onto parvalbumin-expressing interneurons feature larger PSD-95 clusters and both larger and denser Munc13 clusters than excitatory synapses onto other excitatory neurons (Ex → Ex synapses) [10]. These differences likely affect the organization of NMDA and AMPA receptors and the docking and priming of synaptic vesicles at the presynaptic membrane, leading to functional variations and changes in synaptic strength. This suggests that protein cluster organization is a critical factor influencing synaptic properties, supporting the idea that different synapse types have distinct structures to fulfill their specific physiological roles within neural circuits.

In addition to PVINs, we looked to investigate excitatory synapses made onto other inhibitory neuron subtypes (Ex→Inh synapses), identifiable by specific markers (Table 1). Given the diversity of inhibitory neurons and their unique morphological, electrophysiological, and molecular properties, we hypothesized that there would be finer distinctions in synaptic organization among the different types of inhibitory neurons. To explore this concept, we compared the organization of Munc13 and PSD-95 among various synapse subtypes. We utilized five neuropeptide markers (PV, SST, CCK, NPY, and VIP) to identify and differentiate subtypes within Ex→Inh synapses. Additionally, we applied deep learning to impartially determine Ex→Inh synapse subtypes and identify distinguishing synaptic features between them. Our results reveal significant differences between Ex→Inh synapses and Ex→Ex synapses, with Munc13 clusters appearing larger and denser in Ex→Inh synapses. This indicates a strong correlation between postsynaptic cell identity and presynaptic protein organization. Additionally, we observe notable differences among Ex→Inh synapse subtypes, suggesting further differentiation in synaptic characteristics within these subtypes.

TABLE I. TARGETED INHIBITORY NEURON SUBTYPES

<i>Inhibitory Neuron Type(s)</i>	<i>Subtype Marker Used</i>
Axo-axonic	Parvalbumin (PV)
Parvalbumin- expressing basket cells	Parvalbumin (PV)
Bistratified Cells	Neuropeptide Y (NPY) and Somatostatin (SST)
Cholecystokinin-expressing basket cells	Cholecystokinin (CCK)
Dendrite targeting CCK expressing Interneurons	Cholecystokinin (CCK)
Oriens lacunosum-moleculare interneurons	Somatostatin (SST)
Neurogliaform	Neuropeptide Y (NPY)
Ivy cells	Neuropeptide Y (NPY)
Interneuron selective interneurons	Vasoactive intestinal peptide (VIP)

II. RESULTS

A. Inhibitory Subtype Populations

To identify various inhibitory neuron subtypes, we fixed and stained rat hippocampal cultures for NeuN, GAD67, and one of five commonly used markers to identify different classes of inhibitory neurons (Fig. 1a). These markers were used individually or in pairs to label and distinguish specific inhibitory subtypes. Parvalbumin (PV) serves as a marker for PV-expressing neuron subtypes, such as Axo-axonic cells and Parvalbumin-expressing basket cells (Table 1). The neuropeptides Cholecystokinin (CCK), Somatostatin (SST), Neuropeptide Y (NPY), and Vasoactive Intestinal Peptide (VIP) are used as markers for various other subtypes of inhibitory neurons. CCK specifically identifies CCK-expressing basket cells and dendrite-targeting CCK-expressing interneurons, while SST marks Oriens lacunosum-moleculare interneurons. NPY is a marker for neurogliaform and ivy cells, and VIP is specific to interneuron selective interneurons.

To determine which inhibitory neuron subtypes could be further assessed in the synaptic analysis, we examined the proportions of each subtype by using a tile imaging method to capture large areas of each coverslip. A custom ImageJ macro was employed to identify neurons based on the NeuN stain and measure the intensity of GAD67 and the corresponding marker, with positive cells identified as high-intensity outliers. For each marker, we calculated the percentage of marker-positive cells out of the total number of cells, averaging the results from two coverslips each week for three weeks. Our data displayed an average of 10.23% of neurons per coverslip as GAD67 positive (Fig. 1c), aligning with previous research (10) that suggests inhibitory neurons comprise approximately 10% of the total neuronal population in culture.

In parallel to measuring subtype percentages, we analyzed the morphological characteristics of each subtype, utilizing a lentiviral vector (pFDlxBW) that expressed mTagBFP2 under the mDlx1 enhancer to limit expression to inhibitory neurons

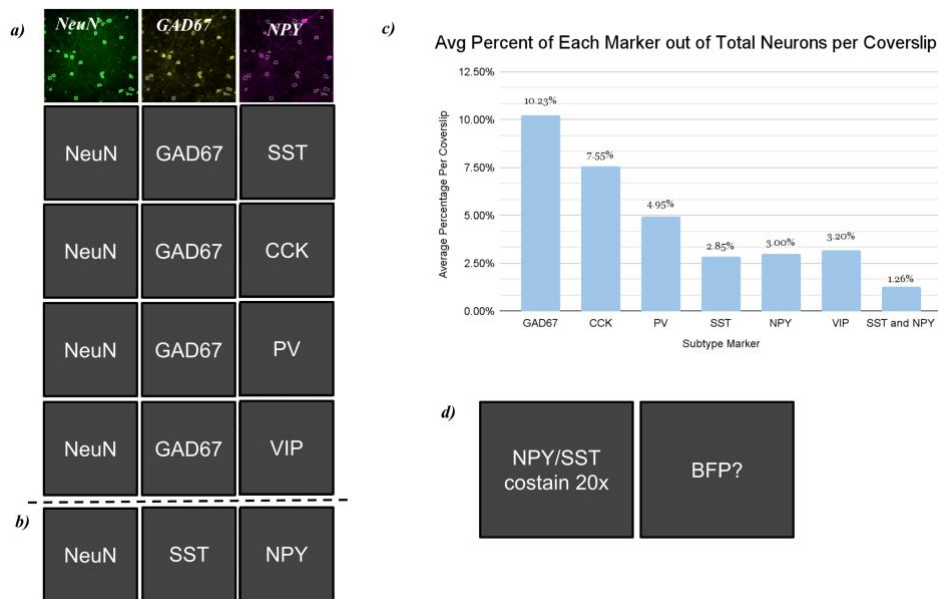


Fig. 1. Example of a Figure caption. (Figure caption)

and visualize the complete neuron. We found that 4.95% of neurons per coverslip were PV-expressing neurons (Fig. 1c) and exhibited larger somas with long, prominent dendrites (Fig. 1a), whereas CCK-expressing neurons had a population of 7.55% and typically displayed smaller somas with short and spindly dendrites. The VIP-expressing neurons had a population of 3.20% and medium-sized somas with thin dendrites of varying lengths. In contrast to the consistency within the PV, CCK, and VIP cell populations, SST-expressing (2.85%) and NPY-expressing (3.00%) neurons each exhibited two distinct morphologies. The first type displayed large somas with thick dendrites, while the second type had smaller somas with thinner and spindly dendrites. Notably, SST and NPY do not express exclusively in cells, and cells that express both markers are an inhibitory subtype called bistratified cells. To understand the impact of bistratified cells on the observed morphological variations, we assessed their proportion in our culture by performing a co-staining of SST and NPY (Fig. 1b). We found that 49.5% of NPY-positive cells also expressed SST and 48.5% of all SST-positive cells also expressed NPY. Additionally, 1.26% of all neurons in our culture exhibited dual expression of both markers, indicating the small yet significant presence of bistratified cells (Fig. 1c). Notably, the neurons that express both SST and NPY appear to be larger with thicker dendrites, suggesting distinct morphological characteristics compared to cells expressing only one of the markers (Fig. 1d).

B. Synaptic Analysis

After observing the proportions of each inhibitory subtype and their unique morphologies, we proceeded with a synaptic analysis to investigate the diversity of synaptic protein content at excitatory synapses onto other excitatory neurons (Ex→Ex) and different inhibitory subtypes (Ex→Inh). To conduct the synaptic analysis, we infected neurons either with the Inh-mTagBFP2 virus or a lentivirus to label excitatory neurons expressing EGFP controlled by a CaMKII promoter. Subsequently, the coverslips were stained for one of the five subtype-specific markers, and all cells were stained for Munc13 and PSD-95 (Fig. 2a). Marker-positive cells were imaged at 60x magnification, and a SynQuant semi-automated

program was utilized to quantify synapses from three independent cultures.

To investigate the influence of postsynaptic cell type on postsynaptic protein organization, we analyzed the characteristics of PSD-95 clusters for each inhibitory neuron subtype. After averaging the data per cell, we found minimal changes in the area of PSD-95 clusters between Ex→Inh and Ex→Ex synapses (Fig. 2b). To further investigate the protein density within these clusters, we measured the integrated intensity and normalized it to the mean value for the Ex→Ex synapses within the same culture. We found no significant difference between Ex→Inh and Ex→Ex synapses (Fig. 2c). These results indicate that PSD-95 clusters maintain relatively consistent size and protein density despite changing the inhibitory neuron subtype, which suggests that PSD-95 organization is relatively stable between subtypes.

We then extended our analysis to examine Munc13 clusters in the presynaptic active zone. In contrast to PSD-95, we observed a significant increase in the size of Munc13 clusters at certain excitatory synapses. Specifically, synapses onto PV, SST, and NPY-expressing inhibitory neurons contained Munc13 clusters of greater area in comparison to Ex→Ex synapses (Fig. 2d). Similarly, the integrated intensity showed that, on average, Munc13 cluster intensity was 1.39 times higher in all Ex→Inh synapses compared to Ex→Ex synapses (Fig. 2e). These results imply that Munc13 clusters are not only larger at most Ex→Inh synapses compared to Ex→Ex synapses but also consistently exhibit greater protein density, regardless of the postsynaptic subtype. Therefore, our findings suggest that Ex→Inh synapses possess a larger and more densely packed presynaptic molecular machinery but retain a relatively stable postsynaptic organization compared to Ex→Ex synapses.

Given their frequent interaction, gaining insights into the relationship between the organization of Munc13 and PSD-95 is essential for comprehending synaptic transmission. To reveal a pattern in this relationship, we examined the ratio of Munc13 to PSD-95 content for each synapse, categorized by subtype. Our analysis revealed significant differences in the ratio between inhibitory neuron subtypes, with SST and VIP

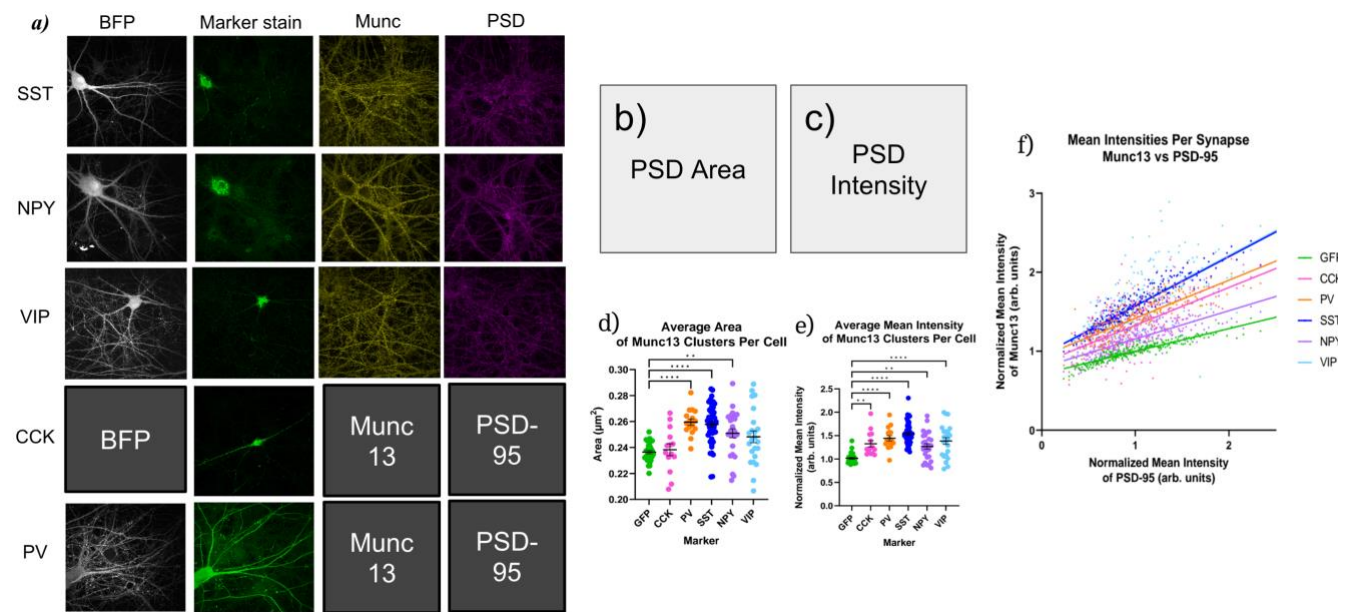


Fig. 1. Example of a Figure caption. (Figure caption)

showing a statistically different trend from PV and CCK (Fig. 2f). Additionally, NPY exhibited a statistically distinct Munc13 to PSD-95 ratio compared to the other subtypes. These findings indicate that each subtype possesses a unique protein organization, which differs significantly from that observed in Ex→Ex synapses. Ultimately, our results indicate larger and denser Munc13 clusters in Ex→Inh synapses compared to Ex→Ex synapses as well as statistically different Munc13 to PSD-95 ratios between inhibitory neuron subtypes, suggesting unique protein organization among those subtypes.

C. Unsupervised Machine Learning

To impartially determine Ex→Inh synapse subtypes, we employed unsupervised machine learning algorithms to cluster synapses based on Munc13 and PSD-95 features. We extracted additional information from the confocal microscopy images, acquiring 20 features for both Munc13 and PSD-95 clusters, resulting in a total of 40 features per synapse. To interpret the data in two dimensions, we performed dimensional reduction of the features using UMAP and rendered multiple UMAPs with each subtype highlighted to visualize the results (Fig. 4a). While the data does not form discrete clusters, noticeable trends in subtype density can still be observed. Specifically, GFP (blue) is greater towards the left hemisphere while SST (red) is most densely located in the right center. Notably, the overlap between the two groups largely drops off with GFP having a sparse population in that area, indicating distinct features between the two subtypes. Additionally, PV demonstrates a gradual increase in density towards the right side of the plot, while CCK, NPY, and VIP show a generally uniform distribution across the plot. Our next step involved using HDBSCAN on the umap output for unsupervised clustering, where we filtered out noise data and calculated subtype percentages within the identified clusters. The results revealed four clusters, each demonstrating diverse subtype distributions (Fig. 4b). Clusters 1 and 2 prominently feature GFP, comprising 51.85% and 51.32%, respectively (Table 4). Cluster 3 exhibits the highest proportion of SST at 48.08% and contains the highest PV percentage among clusters at 17.53%. CCK and NPY are evenly distributed across clusters, ranging from 3.00% to 4.43% and 13.02% to 14.09%, respectively. VIP shows similar dispersion with percentages of 4.67%, 5.91%, and 8.65% in clusters 1, 2, and 3, respectively. Overall, while the data shows overlapping

and/or uniformly distributed subtypes, the underlying gradients in the UMAP and weighted distribution of percentages across clusters suggest the presence of potential unique clusters.

TABLE II. SUBTYPE PERCENTAGE PER CLUSTER

Cluster	GFP	PV	CCK	SST	NPY	VIP
-1	39.98	11.38	4.26	23.15	15.19	6.04
0	8.73	17.53	3.99	48.08	13.02	8.65
1	51.85	6.64	3.00	20.00	13.84	4.67
2	51.32	6.69	4.43	17.35	14.29	5.91

III. DISCUSSION

We examined the proportions, morphology, and synaptic organization of inhibitory neuron subtypes in a cultured system, observing substantial percentages of each subtype in our cultures, with unique morphological characteristics distinguishing them. Interestingly, the postsynaptic organization remained consistent across inhibitory subtypes, as indicated by stable PSD-95 clusters. However, the presynaptic Munc13 clusters at Ex→Inh synapses were larger and more densely packed compared to Ex→Ex synapses, suggesting greater molecular machinery. Furthermore, we identified unique ratios of Munc13 to PSD-95 among inhibitory subtypes, reflecting distinct protein organization patterns. These findings provide valuable information into the relationship between postsynaptic cell type, specifically different inhibitory neuron subtypes, and the organization of synaptic proteins at excitatory synapses onto these inhibitory neurons, laying the foundation for future investigations in this field.

Our findings on Munc13 highlight significant differences in presynaptic protein organization among synapses with varying postsynaptic cell identities, suggesting the characteristics of Munc13 clusters depend on the identity of the postsynaptic cell. This regulation of presynaptic Munc13 clusters and their role in vesicle docking and neurotransmitter release suggests significant functional differences in presynaptic behavior among synapse types. Specifically, all Ex→Inh synapses have larger and denser Munc13 clusters,

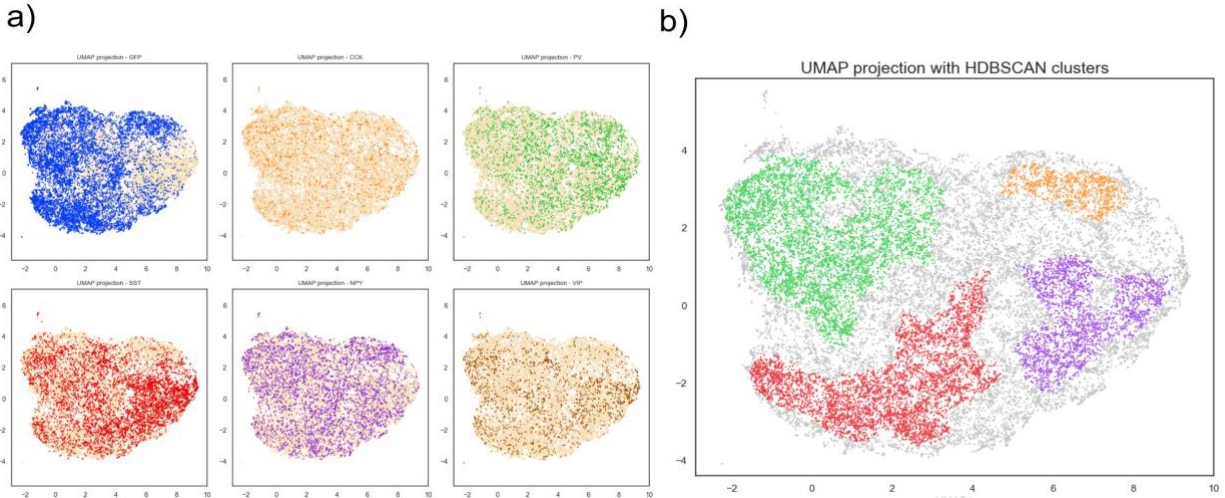


Fig. 1. Example of a Figure caption. (Figure caption)

suggesting an increase in the number of synaptic vesicle release sites and probability of release at all Ex→Inh synapses. In contrast, our analysis of PSD-95 clusters revealed consistent postsynaptic organization across different subtypes, indicating relatively unchanged features. Unlike previous findings that observed an increase in PSD-95 cluster area and intensity [10], we do not observe such increases in our dataset, suggesting that while the presynaptic components exhibit variability, the postsynaptic components maintain a stable structure. Differences in experimental conditions or methodological approaches between our study and previous research might account for these discrepancies. It is also important to note that while the cluster area and intensity was not statistically significant, we still observed an overall increase in PSD-95 clusters. Therefore, increasing the sample size and repeating the experiment might result in a significant increase in PSD-95 clustering. This suggests that further investigation with larger datasets and refined methodologies could provide a clearer understanding of the role of PSD-95 in synaptic organization and its potential variability across different synapse types.

Our results also demonstrate a varying trend in the Munc13 to PSD-95 ratio among Ex→Inh synapse subtypes. Notably, subtypes such as SST and VIP exhibit a similar synaptic organization, which is significantly different from the organization observed in PV and CCK. Additionally, NPY displays a statistically distinct trend compared to all other subtypes. In addition to the variations in the Munc13 to PSD-95 ratio, we can detect trends in the UMAP and distributions in the clusters. We were able to identify four diverse clusters with varying subtype distributions, highlighting potential unique clusters despite overlapping distributions. Specifically, GFP was the most dominant in cluster 1 and 2 and SST was the most dominant in cluster 3 which likely indicates that the two subtypes have the most distinct synaptic characteristic, allowing for differentiation between the two. Due to these significant variations, we hypothesize that each postsynaptic subtype corresponds to a unique set of presynaptic characteristics, forming a synaptic protein architecture that is finely tuned to ensure proper function and fulfill the distinct roles of each subtype within neural circuits. Further investigation of these trends can be achieved via super resolution microscopy that distinguishes organization differences at Ex→Inh synapse types on a nanoscale level. Additionally, supervised machine learning can be utilized to identify specific features that differentiate synapse types.

Certain factors, such as the method for cell type selection, were not ideal due to a significant proportion of cells that express more than one marker (Table 1). A Cre-driver mouse line would allow for cell-specific genetic expression and improved accuracy. Furthermore, variability among CCK neurons can be addressed by increasing the sample size in future experiments (Table 2). The low sample size likely caused the subtype to be underrepresented, potentially influencing the representation of these classes in our machine learning analysis. Additionally, it's important to note that the dimensionality reduction and unsupervised learning analyses were based on data extracted from confocal microscopy images, limiting the number of features extracted from clusters. By utilizing super-resolution imaging techniques such as DNA-PAINT, we can better extract features from synapses and develop more detailed profiles. Acquiring additional features such as the transsynaptic alignment of Munc13 with PSD-95 may provide further insights into

synaptic function and organization. Investigation using super-resolution microscopy could also elucidate nanoscale differences in Munc13 and PSD-95 organization, potentially revealing additional distinctions among synapse types.

Ultimately, our goal is to characterize the protein profiles across various Ex→Inh synapse types and establish connections between protein expression profiles, synaptic structure, and functional diversity throughout the brain. By doing so, we aim to gain a comprehensive understanding of synaptic diversity and the distinct types of synapses in neural circuits. This foundational knowledge will support future experiments designed to elucidate the organization of proteins at synapses. Such research has the potential to identify molecular abnormalities related to synaptic protein clustering, which are often associated with neurocognitive diseases. By understanding these abnormalities, we may uncover new insights into the mechanisms underlying these conditions and pave the way for the development of targeted therapeutic strategies.

TABLE III. SAMPLE SIZE FOR SYNAPTIC ANALYSIS

<i>Subtype Marker</i>	<i>Total Number of Cells Analyzed</i>	<i>Total Number of Synapses Analyzed</i>
GFP	30	11519
CCK	14	1112
PV	16	2769
SST	47	6251
NPY	26	4015
VIP	23	1619

IV. MATERIALS AND METHODS

A. DNA Constructs and Lentivirus

pFCaGW and pFDlxBW are lentiviral vectors derived from pFW that express either EGFP under the CaMKII promoter (“CaG”) to limit expression to excitatory neurons, or mTagBFP2 under the mDlx1 enhancer/minimal human beta-globin promoter (“DlxB”) to limit expression to inhibitory neurons, respectively. pFW and pFCaGW have been previously described [10]. pFDlxBW was assembled by inserting a synthetic DNA fragment based on pAAV-mDlx-GFP-Fishell-1 (Addgene plasmid # 83900) containing the mDlx1/HBB enhancer/promoter and EGFP swapped for mTagBFP2 (Twist Biosciences) into the EcoRI site of pFW with NEB HIFI Assembly. psPAX2 (Addgene plasmid # 12260; <http://n2t.net/addgene:12260>; RRID:Addgene_12260) and pMD2.G (Addgene plasmid #12259; <http://n2t.net/addgene:12259>; RRID:Addgene_12259) were gifts from Didier Trono.

Lentivirus was generated from pFCaGW and pFDlxBW as previously described [10]. Briefly, 5 x 10⁶ HEK293T cells (ATCC CRL-3216) were transfected with 6 µg pFCaGW or pFDlxBW, 4 µg psPAX2, and 2 µg pMD2.G using PEI for 4-6 hours. The virus was harvested from the media 48h later, filtered through a 0.45 µm PES filter, and stored unconcentrated at -80°C in single-use aliquots.

B. Primary Neuron Culture

All animal procedures were approved by the University of Maryland Animal Use and Care Committee. Dissociated

hippocampal neurons cultures were prepared from E18 Sprague-Dawley rat embryos (Charles River) of both sexes as previously described (we'll need a citation) and plated on poly-L-lysine-coated coverslips at 30,000 cells/coverslip (18 mm, #1.5, Warner Instruments). Neurons were maintained in Neurobasal A + GlutaMax, genatmicin, and B27 supplement, and FUDR was added after 1 week to suppress glial growth. Neurons were infected at DIV5-7 with 100 μ l of pFCaGW or pFDlxBW lentivirus and fixed at DIV18-21 for all experiments.

C. Immunostaining

Coverslips infected with pFDlxBW or pFCaGW were fixed with 4% PFA + 4% sucrose in PBS for 10 minutes at room temperature (RT), washed 3 x 5 minutes with PBS + 100 mM glycine (PBSG), then permeabilized for 20 minutes at RT with 0.3% Triton X-100 in PBSG. The coverslips were then incubated with Image-iT™ FX Signal Enhancer (Thermo) for 30 minutes at RT to reduce charge-based non-specific secondary antibody binding in the soma and nucleus, then were blocked for 1 hour at RT with 5% donkey serum + 5% goat serum + 0.2% Triton X-100 in PBSG.

For the measurement of inhibitory neuron subtype proportions, pFDlxBW-infected neurons were stained overnight at 4°C with the primary antibodies mouse anti-GAD67 IgG2B and mouse anti-NeuN IgG1, as well as with one antiserum against an inhibitory neuron subtype marker: guinea pig anti-PV, guinea pig anti-CCK, guinea pig anti-SST, guinea pig anti-NPY, or guinea pig anti-VIP. Antibodies/sera were diluted in 2.5% donkey serum + 2.5% goat serum + 0.1% Triton X-100 in PBSG. On the following day, cells were washed for 3 x 5 minutes with PBSG and incubated with the secondary antibodies donkey anti-mouse IgG2B AlexaFluor488, goat anti-mouse IgG1 Cy3B, and donkey anti-guinea pig AlexaFluor647 for 1 hour at RT in PBSG. The cells were finally washed for 3 x 5 minutes with PBSG before being stored at 4°C until imaging. For SST/NPY double staining, cells were stained with mouse anti-NeuN IgG1 primary antibody and guinea pig anti-SST and mouse IgG2A anti-NPY antisera, then with goat anti-mouse IgG2A AlexaFluor488, goat anti-mouse IgG1 Cy3B, and donkey anti-guinea pig AlexaFluor647 secondaries, and processed the same as above.

For the measurement of synaptic proteins, pFCaGW or pFDlxBW-infected neurons were stained as above, with the primary antibodies mouse anti-PSD-95 IgG2A, rabbit anti-Munc13, and one of guinea pig anti-PV, guinea pig anti-CCK, guinea pig anti-SST, guinea pig anti-NPY, or guinea pig anti-VIP for pFDlxBW-infected cells, or with chicken anti-GFP for pFCaGW-infected cells. Cells were subsequently stained with the secondary antibodies goat anti-mouse IgG2a Cy3B, donkey anti-rabbit AlexaFluor647, and either donkey anti-guinea pig AlexaFluor488 for pFDlxBW-infected cells, or donkey anti-chicken AlexaFluor488 for pFCaGW-infected cells.

Primary and secondary antibody dilutions are detailed in the table below. Cy3B-conjugated secondaries were generated in-house as previously described [10].

Primary	Source	RRID	Stock Concentration	Dilution
Monoclonal mouse anti-GAD67 IgG2B (clone 126G12)	Synaptic Systems 198211	AB_2721099	0.5 mg/mL	1:500
Monoclonal mouse anti-NeuN IgG1 (clone A60)	Millipore MAB377	AB_2298772	1 mg/mL	1:1000
Guinea pig anti-Parvalbumin (PV) antiserum	Swant GP72	AB_2665495	n/a	1:250
Guinea pig anti-Somatostatin-28 (SST) antiserum	Synaptic Systems 366004	AB_2620126	n/a	1:250
Guinea pig anti-Neuropeptide Y (NPY) antiserum	Synaptic Systems 394004	AB_2721083	n/a	1:250
Guinea pig anti-Cholecystokinin-8 (CCK) antiserum	Synaptic Systems 438004	AB_28144938	n/a	1:250
Guinea pig anti-Vasointestinal Peptide (VIP) antiserum	Synaptic Systems 443005	AB_2832228	n/a	1:100
Monoclonal mouse anti-NPY IgG2A antiserum (clone L115_13)	Gift of James Trimmer	n/a	n/a	1:50
Monoclonal mouse anti-PSD-95 IgG2A (clone K28/43)	Neuromab 75-028	AB_2877189	0.5 mg/mL	1:80
Polyclonal rabbit anti-Munc13-1	Synaptic Systems 126103	AB_887733	0.5 mg/mL	1:250
Chicken anti-GFP	Thermo Fisher A10262	AB_2534023	1 mg/mL	1:1000

Secondary	Source	RRID	Stock concentration	Dilution
Goat anti-mouse IgG2B AlexaFluor488	Thermo Fisher A21141	AB_2535778	2 mg/mL	1:500
Goat anti-mouse IgG1 Cy3B	This work	n/a	NEED TO LOOK UP	1:500
Donkey anti-guinea pig AlexaFluor647	Jackson Immnore search 706-605-148	AB_2340476	1.25 mg/ml	1:500
Goat anti-mouse IgG2A AlexaFluor488	Jackson Immnore search 115-545-206	AB_2338855	1.25 mg/mL	1:500
Goat anti-mouse IgG2A Cy3B	This work	n/a	NEED TO LOOK UP	1:500
Donkey anti-rabbit AlexaFluor647	Jackson Immnore search 711-605-152	AB_2492288	1.25 mg/mL	1:500
Donkey anti-guinea pig AlexaFluor488	Jackson Immnore search 706-545-148	AB_2340472	1.25 mg/mL	1:500
Donkey anti-chicken AlexaFluor488	Jackson Immnore search 703-545-155	AB_2340375	1.25 mg/mL	1:500

D. Confocal Microscopy

All confocal images were acquired on a Nikon TI2 Eclipse inverted microscope with Nikon Plan Apo λ 20x/0.75 NA and Plan Apo λ D 60x/1.42 NA objectives and a Dragonfly confocal unit (Andor). Excitation light (405, 488, 561, or 638 nm) from an Andor ILE was flattened by a beam conditioning unit (Andor) and passed to the sample by a 405/488/561/640 polychroic (Chroma). Emission light was passed through an

appropriate bandpass filter (Em01-R442/647 (Semrock), ET525/50, or ET600/50 (Chroma)) and collected on a Zyla 4.2 sCMOS camera (Andor). For subtype percentages, cells were imaged with the 20x objective in widefield mode at 20% laser power and 200 ms exposure, and 10 x 10 tiled images were collected and stitched using Fusion (Andor). The focal plane was maintained using Nikon's Perfect Focus system. Synaptic proteins were imaged as described previously [10]. Cells of interest were imaged with the 60x objective in 10 μm confocal z-stacks with 0.3 μm z-steps at 50% laser power ($\sim 1\text{-}2\text{ W/cm}^2$) with 200 ms exposure. Reference bead stacks for chromatic aberration correction were acquired by immobilizing 100 nm TetraSpeck beads (Invitrogen) on coverslips prepared as for neuron culture, and imaging z-stacks as above.

E. Confocal Analysis

All experimental z-stacks were converted to maximum intensity projections in FIJI [11]. To analyze subtype proportions, we programmed a custom macro in FIJI to open each image, use an auto threshold to select cells, count the number of cells, and measure the intensities of both GAD67 and the specific markers within the cell border. We ran a flat field correction on each image by stacking many NeuN images together and creating a pseudo flat field image where the signal is approximately even. Then we ran an operation that would divide our selected image by the pseudo flatfield image, creating a flattened image that would be further filtered. We chose to use a median blur to smooth the NeuN signal and ran a series of commands that would fill holes and separate any cells that were touching. For each image, the macro ran the flat field operation, filtered the NeuN image, selected particles greater than 150 square microns (μm^2), measured the intensity of each channel within the selected cell, and saved the data as an Excel file. After running the macro on all our images, we had 1500 files that each held data for one channel on one image. Outliers were removed from the data set and values were averaged per cell.

For the synaptic protein cluster analysis, each image was cropped and manually thresholded to select a clean dendrite segment. The image was smoothed using the dilate function, followed by a close holes function. In addition, small background particles in the mask were removed, and the resulting image was used as the dendrite mask. SynQuant was then used to identify presynaptic puncta that overlapped with postsynaptic puncta. Any presynaptic puncta that were not adjacent to a respective postsynaptic puncta were discarded. Similarly, any postsynaptic puncta that were not adjacent to a respective presynaptic puncta were discarded. This ensured that each puncta was paired with the proper and most proximate puncta. Finally, measurements of PSD-95 and Munc13 puncta were taken, outliers were excluded, and the intensities were normalized to the mean of the Ex $\rightarrow$ Ex group per week to account for staining and imaging variations across culture weeks.

F. Dimensional Reduction and Clustering

To provide comprehensive information for synapse classification, we extracted additional features from the confocal microscopy images. For each synapse, we acquired 32 features for both Munc13 and PSD-95 using ImageJ. To interpret the data in two dimensions, we employed uniform manifold approximation and projection (UMAP) for

dimensionality reduction [12]. Using the Python implementation of umap-learn with optimized parameters ($n_neighbors = 5$, $min_dist = 0.0$, $n_components = 2$, and $metric = 'euclidean'$), we reduced the 64 features to a two-dimensional form. For visualization of the output, we rendered multiple UMAPs with each subtype highlighted. For unsupervised clustering, we implemented the Hierarchical density-based spatial clustering of applications with noise (HDBSCAN) on the umap output using the hdbscan module from python with $min_cluster_size = 380$. The percentages of each subtype present in each resulting cluster were calculated, and a heatmap representing the average value of each feature per cluster was generated.

ACKNOWLEDGMENT

We thank the members of the Blanpied Laboratory for helpful discussions and technical support.

REFERENCES

- [1] O'Rourke, N., Weiler, N., Micheva, K. et al. Deep molecular diversity of mammalian synapses: why it matters and how to measure it. *Nat Rev Neurosci* 13, 365–379 (2012). <https://doi.org/10.1038/nrn3170>
- [2] Volk, L., Chiu, S. L., Sharma, K., & Huganir, R. L. (2015). Glutamate synapses in human cognitive disorders. *Annual review of neuroscience*, 38, 127–149. <https://doi.org/10.1146/annurev-neuro-071714-033821>
- [3] Bayés, A., van de Lagemaat, L. N., Collins, M. O., Croning, M. D., Whittle, I. R., Choudhary, J. S., & Grant, S. G. (2011). Characterization of the proteome, diseases and evolution of the human postsynaptic density. *Nature neuroscience*, 14(1), 19–21. <https://doi.org/10.1038/nn.2719>
- [4] Südhof T. C. (2008). Neuroligins and neuroligins link synaptic function to cognitive disease. *Nature*, 455(7215), 903–911. <https://doi.org/10.1038/nature07456>
- [5] Karlocai, M. R., Heredi, J., Benedek, T., Holderith, N., Lorincz, A., & Nusser, Z. (2021). Variability in the Munc13-1 content of excitatory release sites. *eLife*, 10, e67468. <https://doi.org/10.7554/eLife.67468>
- [6] Sakamoto, H., Ariyoshi, T., Kimpara, N., Sugao, K., Taiko, I., Takikawa, K., Asanuma, D., Namiki, S., & Hirose, K. (2018). Synaptic weight set by Munc13-1 supramolecular assemblies. *Nature neuroscience*, 21(1), 41–49. <https://doi.org/10.1038/s41593-017-0041-9>
- [7] Fukata, Y., Dimitrov, A., Boncompain, G., Vielemeyer, O., Perez, F., & Fukata, M. (2013). Local palmitoylation cycles define activity-regulated postsynaptic subdomains. *The Journal of cell biology*, 202(1), 145–161. <https://doi.org/10.1083/jcb.201302071>
- [8] Nair, D., Hosy, E., Petersen, J. D., Constals, A., Giannone, G., Choquet, D., & Sibarita, J. B. (2013). Super-resolution imaging reveals that AMPA receptors inside synapses are dynamically organized in nanodomains regulated by PSD95. *The Journal of neuroscience : the official journal of the Society for Neuroscience*, 33(32), 13204–13224. <https://doi.org/10.1523/JNEUROSCI.2381-12.2013>
- [9] Danielson, E., Perez de Arce, K., Cimini, B., Wamhoff, E. C., Singh, S., Cottrell, J. R., Carpenter, A. E., & Bathe, M. (2021). Molecular Diversity of Glutamatergic and GABAergic Synapses from Multiplexed Fluorescence Imaging. *eNeuro*, 8(1), ENEURO.0286-20.2020. <https://doi.org/10.1523/ENEURO.0286-20.2020>
- [10] Dharmasri, P. A., Levy, A. D., & Blanpied, T. A. (2024). Differential nanoscale organization of excitatory synapses onto excitatory vs. inhibitory neurons. *Proceedings of the National Academy of Sciences of the United States of America*, 121(17), e2315379121. <https://doi.org/10.1073/pnas.2315379121>
- [11] Schindelin, J., Arganda-Carreras, I., Frise, E., Kaynig, V., Longair, M., Pietzsch, T., Preibisch, S., Rueden, C., Saalfeld, S., Schmid, B., Tinevez, J. Y., White, D. J., Hartenstein, V., Eliceiri, K., Tomancak, P., & Cardona, A. (2012). Fiji: an open-source platform for biological-image analysis. *Nature methods*, 9(7), 676–682. <https://doi.org/10.1038/nmeth.2019>
- [12] McInnes, L., Healy, J., & Melville, J. (2018). Umap: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426*.